

Industrial Internship Report on " Smart Home Automation System Using IoT"

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt. Ltd. (UCT).

This internship was focused on a project/problem statement provided by UCT. During the internship period, I worked on the project titled "Smart Home Automation System Using IoT".

The project was developed using an ESP8266 NodeMCU microcontroller, DHT11 temperature and humidity sensor, and a relay module. The system enables monitoring of environmental conditions and remote control of household appliances such as lights and fans through Wi-Fi connectivity. The project demonstrates the practical implementation of Embedded Systems and Internet of Things (IoT) technologies in home automation applications.

This internship gave me a valuable opportunity to gain exposure to industry-relevant technologies and practical problem-solving approaches. During the internship, I enhanced my knowledge of Embedded Systems, IoT, sensor interfacing, wireless communication, embedded programming, circuit design, troubleshooting, and project documentation.

Overall, it was a rewarding learning experience that improved my technical skills and provided practical exposure to real-world applications of Embedded Systems and IoT technologies.

TABLE OF CONTENTS

1	Preface	3
2	Introduction	3
2.1	About UniConverge Technologies Pvt Ltd	4
2.2	About upskill Campus	8
2.3	Objective	10
2.4	Reference	10
2.5	Glossary	10
3	Problem Statement	11
4	Existing and Proposed solution	12
5	Proposed Design/ Model	14
5.1	High Level Diagram (if applicable)	14
5.2	Low Level Diagram (if applicable)	14
5.3	Interfaces (if applicable)	15
6	Performance Test	16
6.1	Test Plan/ Test Cases	16
6.2	Test Procedure	17
6.3	Performance Outcome	17
7	My Learnings	18
8	Future Work Scope	19

1 Preface

The Free Summer Internship in Embedded Systems and IoT organized by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT) provided me with an excellent opportunity to enhance my technical knowledge and practical understanding of Embedded Systems and Internet of Things (IoT).

During this internship, I worked on the project “Smart Home Automation System Using IoT”. The project focuses on monitoring environmental conditions and controlling household appliances using an ESP8266 NodeMCU, DHT11 sensor, and relay module. The system demonstrates how IoT technology can be used to automate and remotely monitor home appliances efficiently.

The internship was structured over several weeks, beginning with learning the fundamentals of Embedded Systems, IoT architecture, sensors, microcontrollers, communication protocols, and automation concepts. As the internship progressed, I explored sensor interfacing, Wi-Fi communication, IoT deployment concepts, embedded programming, circuit design, troubleshooting techniques, and system testing methodologies.

This internship helped me understand how industrial projects are planned, developed, tested, and documented. It also improved my problem-solving ability, analytical thinking, debugging skills, and understanding of real-world IoT applications.

I would like to express my sincere gratitude to upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd. (UCT), and all mentors who provided valuable guidance and learning resources throughout the internship. Their support helped me successfully complete the project and gain practical exposure to industry-relevant technologies.

I would also like to encourage my juniors and peers to actively participate in internships and practical projects, as they provide valuable experience, improve technical skills, and help bridge the gap between academic learning and industry requirements.

Overall, this internship was a rewarding learning experience that strengthened my knowledge of Embedded Systems and IoT while preparing me for future technical and professional challenges.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



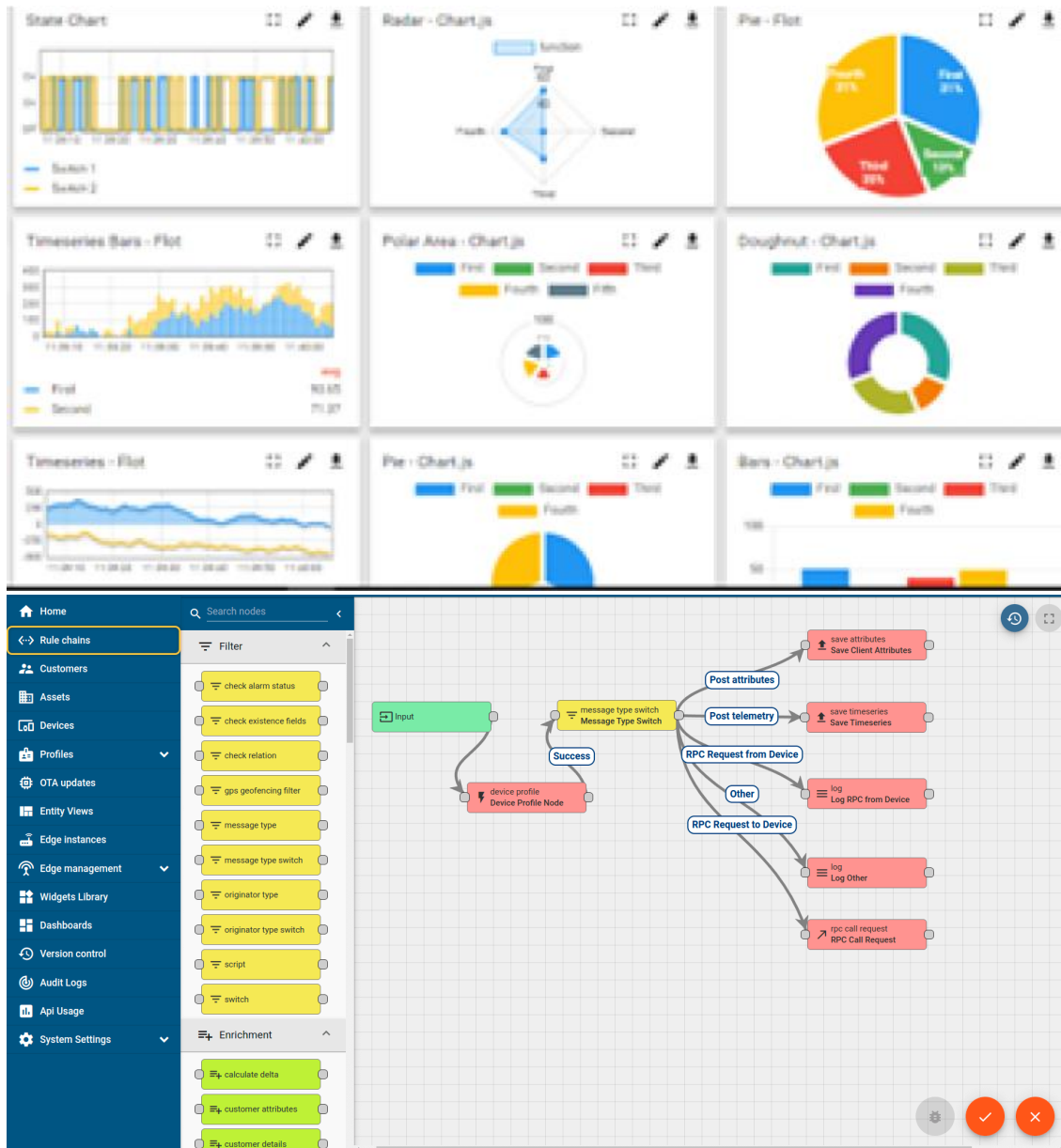
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



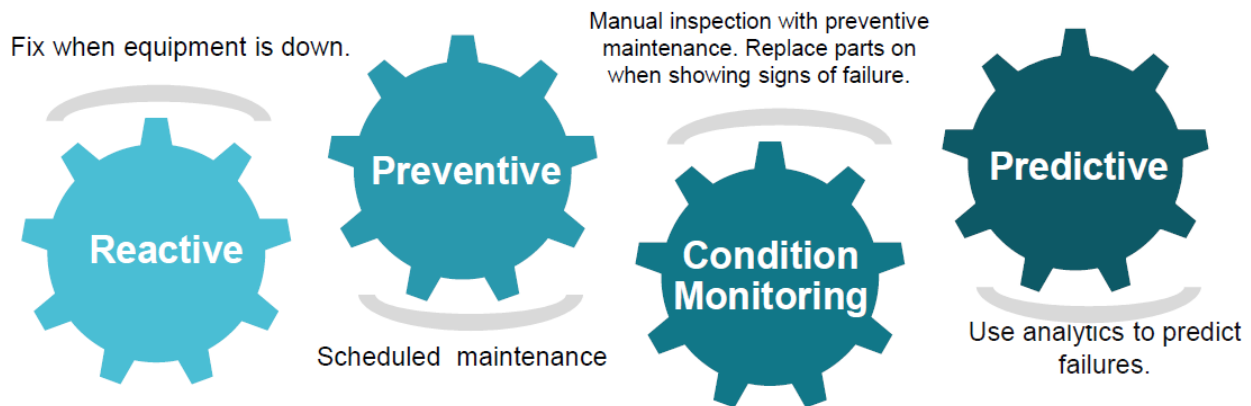


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRaWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

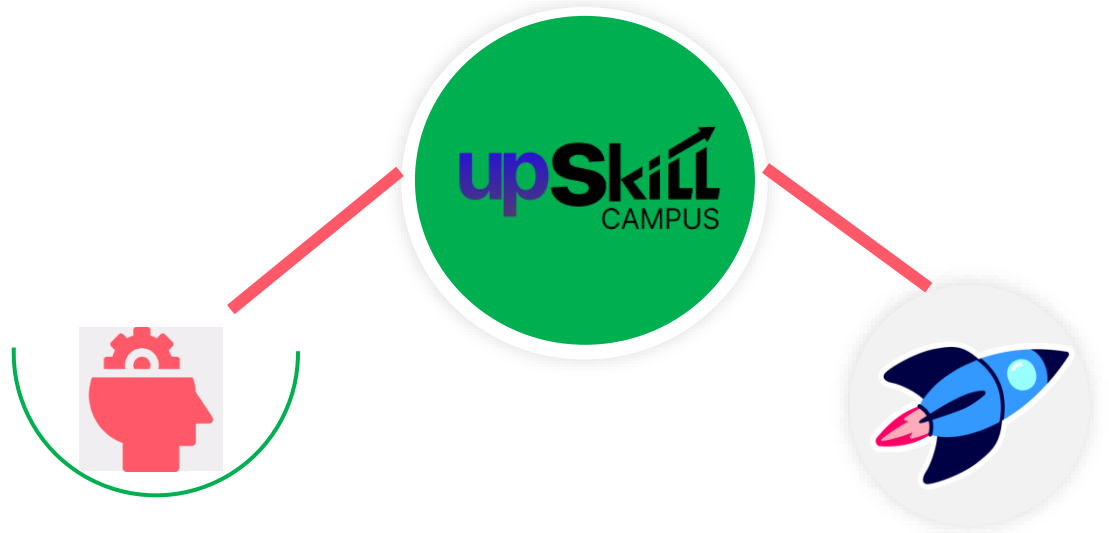
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

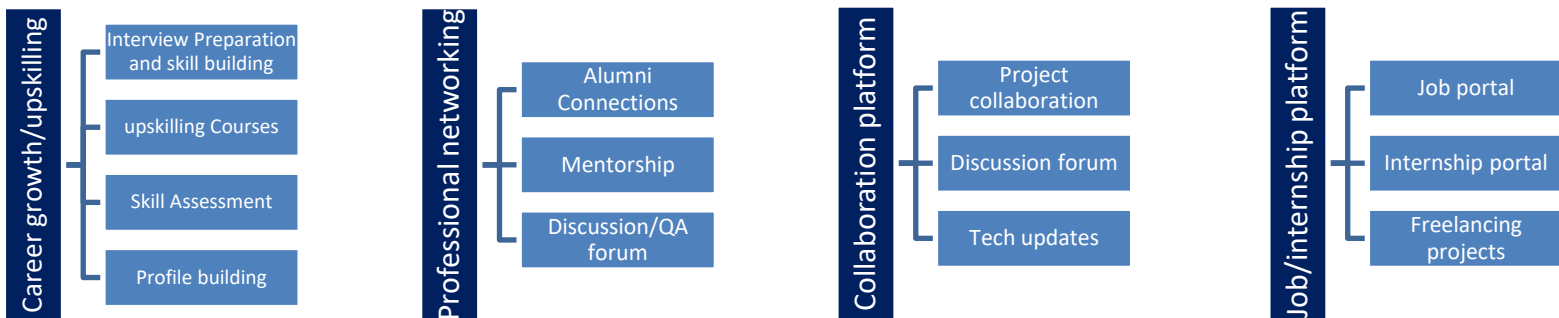
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] ESP8266 NodeMCU Documentation and Arduino IDE Resources.
- [2] DHT11 Temperature and Humidity Sensor Datasheet.
- [3] IoT Learning Materials and Tutorials provided by The IoT Academy and upskill Campus.

2.6 Glossary

Terms	Acronym
Internet of Things	IoT
Temperature and Humidity Sensor	DHT11
Microcontroller Unit	MCU
Wireless Fidelity	Wi-Fi
General Purpose Input Output	GPIO
Hypertext Transfer Protocol	HTTP
Integrated Development Environment	IDE

3 Problem Statement

With the rapid growth of Internet of Things (IoT) technologies, there is an increasing demand for smart systems that improve convenience, efficiency, and automation in daily life. Traditional home electrical systems mainly rely on manual operation, where users must physically switch appliances such as lights and fans on or off. This conventional approach lacks flexibility, remote accessibility, and intelligent monitoring capabilities.

In modern lifestyles, people often require the ability to monitor and control household appliances even when they are away from home. Additionally, the lack of environmental monitoring can lead to inefficient energy usage and reduced user comfort. As a result, there is a growing need for affordable and efficient home automation systems that can provide remote access, real-time monitoring, and automation features.

The primary objective of this project is to design and implement a Smart Home Automation System Using IoT. The system utilizes an ESP8266 NodeMCU microcontroller as the main processing unit, a DHT11 temperature and humidity sensor for environmental monitoring, and a 2-channel relay module for controlling household appliances such as lights and fans. Through Wi-Fi connectivity, the system enables communication between the user and the connected devices.

The proposed solution allows users to monitor temperature and humidity data in real time and remotely control connected appliances through a web-based interface. The relay module acts as an electronic switching mechanism that controls electrical loads, while the ESP8266 provides wireless connectivity and data processing capabilities. The integration of these components creates a simple yet effective IoT-based home automation system.

The project addresses several challenges associated with traditional home automation methods, including the absence of remote control, limited monitoring capabilities, and inefficient energy utilization. By implementing IoT technology, the system enhances user convenience, improves accessibility, and demonstrates how embedded systems can be integrated with wireless communication technologies to develop smart applications.

Furthermore, this project serves as a practical learning platform for understanding key concepts of Embedded Systems and IoT, including sensor interfacing, microcontroller programming, Wi-Fi communication, relay control, system integration, and troubleshooting. The knowledge gained through the development of this project can be applied to more advanced smart home and industrial automation systems in the future.

Overall, the Smart Home Automation System Using IoT provides a cost-effective and scalable solution for modern home automation requirements while demonstrating the practical implementation of IoT technologies in real-world applications.

4 Existing and Proposed solution

Existing Solution

Traditional home electrical systems rely on manual operation of appliances such as lights and fans through physical switches. These systems are simple and commonly used, but they do not provide remote access, monitoring, or automation features. Users must be physically present to control appliances, which can be inconvenient and may lead to inefficient energy usage.

Conventional systems also lack the ability to monitor environmental conditions such as temperature and humidity. As a result, appliance operation is completely dependent on manual control, reducing flexibility and efficiency.

Limitations of Existing Solution

- No remote monitoring or control.
- Lack of automation capabilities.
- Increased energy consumption due to manual operation.
- No real-time environmental monitoring.
- Limited convenience for users.
- Difficult to manage appliances when away from home.

Proposed Solution

The proposed Smart Home Automation System Using IoT is designed to overcome the limitations of traditional home electrical systems. The system uses an ESP8266 NodeMCU microcontroller, DHT11 temperature and humidity sensor, and a 2-channel relay module to provide smart monitoring and control of household appliances.

The ESP8266 microcontroller enables Wi-Fi connectivity, allowing users to access and control appliances remotely through a web-based interface. The DHT11 sensor continuously monitors temperature and humidity conditions, while the relay module controls appliances such as lights and fans.

The system provides real-time monitoring, remote accessibility, and improved convenience. It demonstrates the practical implementation of IoT technology in home

automation and helps users manage appliances more efficiently.

- Advantages of Proposed Solution
- Remote monitoring and control of appliances.
- Real-time temperature and humidity monitoring.
- Improved user convenience and accessibility.
- Better energy management.
- Low-cost and scalable implementation.
- Easy integration with future IoT applications.

4.1 Code Submission (GitHub Link)

<https://github.com/PavanCh26/upskillcampus/blob/main/SmartHomeAutomationSystem.ino>

4.2 Report Submission (GitHub Link)

<https://github.com/PavanCh26/upskillcampus/blob/main/SmartHomeAutomationSystem Pavan USC UCT.pdf>

5 Proposed Design/ Model

The proposed Smart Home Automation System Using IoT is designed to provide remote monitoring and control of household appliances using IoT technology. The system consists of an ESP8266 NodeMCU microcontroller, DHT11 temperature and humidity sensor, and a 2-channel relay module. The ESP8266 acts as the central controller, collecting sensor data and communicating with users through a Wi-Fi network. The relay module is used to control appliances such as lights and fans, while the DHT11 sensor continuously monitors environmental conditions. The overall design ensures efficient communication, real-time monitoring, and convenient appliance control through a web-based interface.

5.1 High Level Diagram (if applicable)

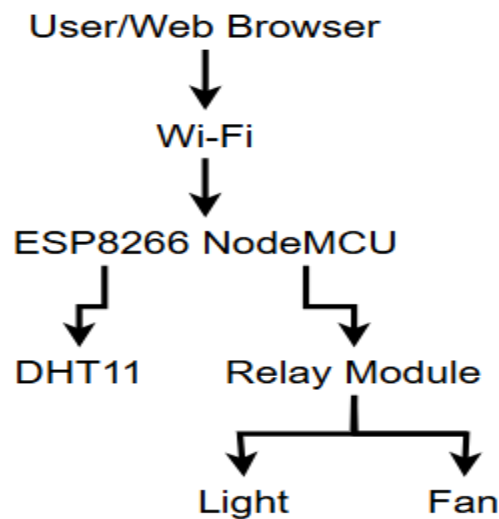
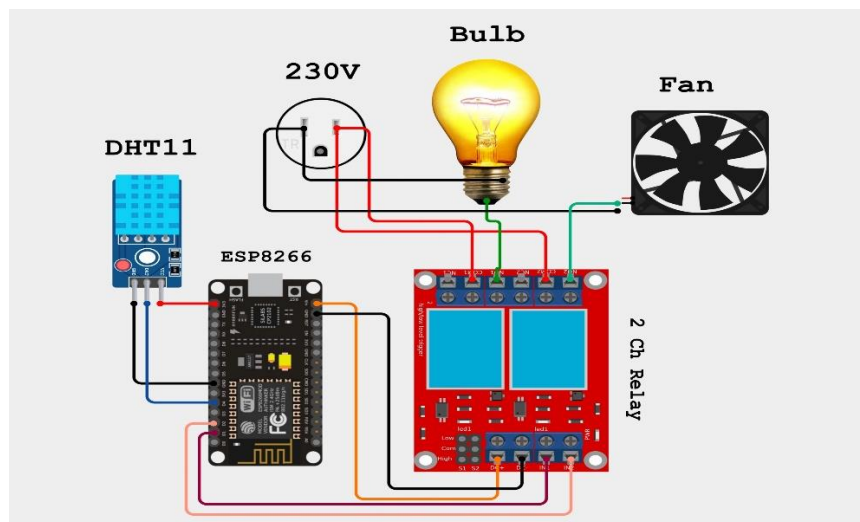


Figure 1: Simple Architecture Diagram

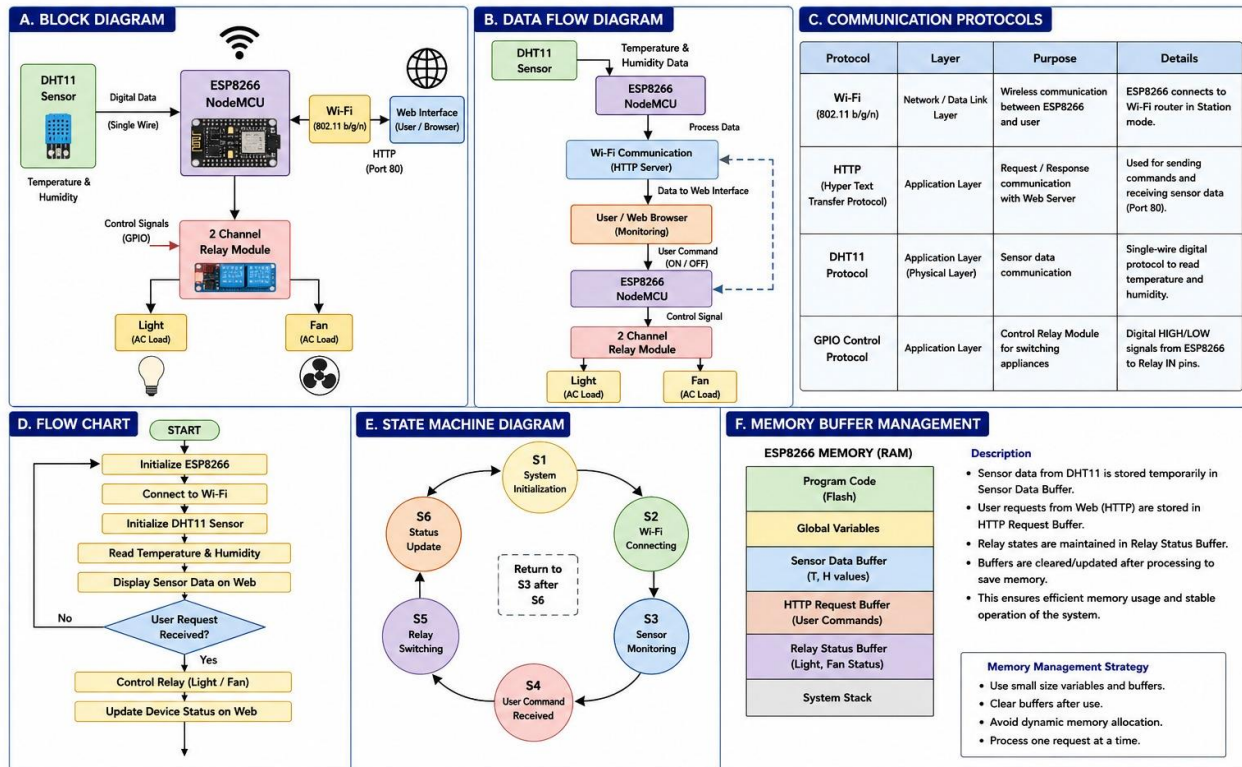
5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

The interface design of the Smart Home Automation System Using IoT illustrates the interaction between hardware and software components. It includes the system block diagram, data flow diagram, communication protocols, flow chart, state machine, and memory buffer management. These interfaces define the communication process between the ESP8266 NodeMCU, DHT11 sensor, relay module, and user interface, ensuring efficient monitoring and control of connected appliances.

The interface design of the Smart Home Automation System Using IoT includes block diagram, data flow, communication protocols, flow chart, state machine and memory buffer management.



6 Performance Test

The Smart Home Automation System Using IoT was evaluated based on key constraints commonly encountered in industrial embedded systems such as memory utilization, communication reliability, response time, sensor accuracy, and power consumption. The ESP8266 NodeMCU provides sufficient memory and processing capability for handling sensor data acquisition, web server communication, and relay control operations. To minimize memory usage, the system processes sensor readings only when required and avoids storing unnecessary data. Communication reliability was considered by utilizing Wi-Fi connectivity and HTTP-based communication between the ESP8266 and the user interface. Multiple tests confirmed stable device control and successful transmission of sensor data under normal operating conditions.

Sensor accuracy was evaluated using the DHT11 temperature and humidity sensor. While the DHT11 provides acceptable readings for basic monitoring applications, more accurate sensors such as DHT22 can be used in future implementations where higher precision is required. Power consumption was minimized through the use of lightweight embedded hardware and efficient control logic. The relay module successfully controlled the connected light and fan without affecting overall system stability. The test results demonstrated reliable operation of the system with stable communication, accurate appliance control, and efficient resource utilization. Future improvements may include enhanced security, cloud integration, and the use of higher-accuracy sensors for industrial-scale applications.

6.1 Test Plan/ Test Cases

Test Case ID	Test Description	Expected Result	Status
TC-01	Power ON ESP8266 NodeMCU	System starts successfully	Pass
TC-02	Connect ESP8266 to Wi-Fi	Wi-Fi connection established	Pass
TC-03	Read DHT11 sensor data	Temperature and humidity displayed	Pass
TC-04	Turn ON Light through web interface	Light switches ON	Pass
TC-05	Turn OFF Light through web interface	Light switches OFF	Pass
TC-06	Turn ON Fan through web interface	Fan switches ON	Pass
TC-07	Turn OFF Fan through web interface	Fan switches OFF	Pass
TC-08	Monitor system continuously	Stable operation observed	Pass

6.2 Test Procedure

The Smart Home Automation System was tested by powering the ESP8266 NodeMCU and connecting it to a Wi-Fi network. The DHT11 sensor was interfaced with the controller to measure temperature and humidity values. The relay module was connected to control the light and fan. After successful initialization, the web interface was accessed through a browser. Various test cases were performed to verify sensor monitoring, appliance control, Wi-Fi communication, and system responsiveness. Multiple ON/OFF operations were executed to ensure reliable functioning of the relay module and connected appliances. The performance of the system was observed under normal operating conditions, and the outputs were verified with the expected results.

6.3 Performance Outcome

The Smart Home Automation System Using IoT successfully performed all intended operations. The ESP8266 NodeMCU established a stable Wi-Fi connection and communicated effectively with the web interface. The DHT11 sensor provided temperature and humidity readings, while the relay module successfully controlled the connected light and fan. The system demonstrated reliable appliance control, real-time monitoring, and efficient communication between hardware and software components. Overall, the project achieved its objectives and validated the practical implementation of IoT-based home automation technology.

7 My learnings

The Free Summer Internship in Embedded Systems and IoT provided me with an excellent opportunity to enhance both my theoretical knowledge and practical skills in the field of Embedded Systems and Internet of Things (IoT). Throughout the internship, I explored various concepts related to microcontrollers, sensors, communication protocols, embedded programming, circuit design, and IoT applications. One of the most valuable aspects of this internship was the opportunity to work on the Smart Home Automation System Using IoT project. Through this project, I learned how different hardware components such as the ESP8266 NodeMCU, DHT11 temperature and humidity sensor, and relay module can be integrated to develop a functional IoT-based automation system. This helped me understand the practical implementation of concepts that are often studied only theoretically in academic courses. During the project development process, I gained hands-on experience in embedded programming using the Arduino IDE. I learned how to write, modify, and debug programs for controlling sensors and actuators. I also improved my understanding of GPIO pin configuration, sensor interfacing techniques, relay control mechanisms, and serial communication methods.

The internship further enhanced my knowledge of IoT communication and wireless networking. I learned how Wi-Fi-enabled microcontrollers communicate with users through web-based interfaces and how real-time monitoring and remote control can be achieved using IoT technologies. This experience helped me understand the importance of connectivity, communication protocols, and system integration in modern automation systems. Another important learning outcome was the development of troubleshooting and debugging skills. During project implementation, I encountered several challenges related to hardware connections, sensor readings, communication stability, and software logic. By identifying the root causes of these issues and applying suitable solutions, I improved my analytical thinking and problem-solving abilities. I also gained valuable experience in preparing technical documentation, weekly progress reports, project reports, and maintaining project files using GitHub. These activities helped me understand the importance of proper documentation and version control in professional software and hardware development environments. Furthermore, the internship improved my self-learning ability and encouraged me to explore new technologies independently. The learning resources, technical materials, and project activities provided a clear understanding of how Embedded Systems and IoT technologies are applied in real-world industrial applications.

Overall, this internship was a highly beneficial learning experience that strengthened my technical knowledge, practical implementation skills, and professional development. The knowledge and experience gained during this internship will support my future academic projects, technical certifications, and career opportunities in the fields of Embedded Systems, IoT, and Automation Technologies.

8 Future work scope

The Smart Home Automation System Using IoT developed during this internship successfully demonstrates the integration of Embedded Systems and IoT technologies for monitoring and controlling household appliances. Although the current system achieves its primary objectives, several enhancements can be implemented in the future to improve functionality, scalability, security, and user experience.

One possible improvement is the integration of cloud platforms for storing and analyzing sensor data. Cloud connectivity would enable users to access historical data, generate reports, and monitor their home environment from any location through the internet. This would make the system more suitable for large-scale IoT applications.

The project can also be enhanced by incorporating additional sensors such as motion sensors, gas sensors, smoke detectors, and light sensors. These sensors would increase the automation capabilities of the system and improve safety and energy efficiency.

Another important area for future development is mobile application integration. A dedicated Android or iOS application could provide a more user-friendly interface for monitoring sensor readings and controlling appliances remotely. Push notifications and real-time alerts could also be implemented to improve user awareness and system responsiveness.

Security can be further improved by implementing authentication mechanisms, encrypted communication protocols, and secure cloud connectivity. These enhancements would protect the system from unauthorized access and improve data privacy.

The current project uses a DHT11 sensor, which is suitable for basic monitoring applications. In future implementations, more accurate sensors such as DHT22 or industrial-grade sensors can be used to improve measurement precision and reliability.

Artificial Intelligence (AI) and Machine Learning (ML) techniques can also be integrated to develop intelligent automation features. The system could automatically control appliances based on user behavior, environmental conditions, and predictive analysis, thereby increasing efficiency and convenience.

Overall, the Smart Home Automation System Using IoT provides a strong foundation for future enhancements and can be expanded into a more advanced smart home solution with improved automation, security, monitoring, and intelligent decision-making capabilities.